Laboratory Tasks

Task 1: String Length and Reversal

```c
#include <stdio.h>
#include <string.h>

int main () {
char s[100];

printf("enter a string: ");
scanf("%s", &s);

printf("length of the string is: %d\n", strlen(s));

int l = 0;
int r = strlen(s)-1;
char t;

while (l < r) {
        t = s[l];
s[l] = s[r];
s[r]=t;
l++;
r--;

}

printf("the reversed string is: %s", s);
return 0;
}
```

```
enter a string: helloaaaa
length of the string is: 9
the reversed string is: aaaaolleh
Process returned 0 (0x0)   execution time : 4.432 s
Press any key to continue.
```

Task 2 : Concatenation and Comparison

```c
#include <stdio.h>
#include <string.h>

int main () {
char s1[100];
char s2[100];

printf("enter both strings: \n");
scanf("%s%s", &s1, &s2);

char s3[200];
strcpy(s3, s1);
strcat(s3, s2);

printf("the new string is: %s\n", s3);

if (strcmp(s1, s2) == 0) {
    printf("the strings are equal\n");
}

else printf("the strings are not equal");
return 0;
}
```

```
enter both strings:
good
apple
the new string is: goodapple
the strings are not equal
Process returned 0 (0x0)   execution time : 6.379 s
Press any key to continue.
```

Task 3 : Palindrome and Vowel/Consonant Count

```c
#include <stdio.h>
#include <string.h>

int main() {

char s[100];
printf("enter a string: \n");
scanf("%s", &s);

int b = 1;

for (int i = 0; i < strlen(s)/2; i++) {
    if (s[i] != s[strlen(s)-i-1]) {
        b = 0;
        break;
        }
}
int v = 0;

for(int i = 0; i < strlen(s); i++) {
    if (s[i] == 'a' || s[i] == 'e' || s[i] == 'i' || s[i] == 'o' || s[i] == 'u') {
    v++;
    }
}

if (b) {
    printf("the string is a palindrome\n");
}

else printf("the string is not a palindrome\n");
printf("the number of vowels is %d and the number of consonants is %d", v, strlen(s)-v);
return 0;
}
```

```
enter a string:
goodmorning
the string is not a palindrome
the number of vowels is 4 and the number of consonants is 7
Process returned 0 (0x0)   execution time : 7.345 s
Press any key to continue.
```

Additional Practice Tasks

Task 1: No. of Words in a Sentence

```c
#include <stdio.h>
#include <string.h>

int main () {

char s[200];
printf("enter a sentence: ");
fgets(s, sizeof(s), stdin);

int a = 0;

for (int i = 0; i < strlen(s); i++) {
        if (s[i] == ' ') {
            a++;
        }
}

printf("the number of words is: %d", a+1);
return 0;
}
```

```
enter a sentence: the quick brown fox jumps over the lazy dog
the number of words is: 9
Process returned 0 (0x0)   execution time : 6.458 s
Press any key to continue.
```

Task 2: Counting of a Specific Character

```c
#include <stdio.h>
#include <string.h>

int main () {

char s[100];
printf("enter a string: ");
scanf("%s", s);

char a;
printf("enter the character you want to count: ");
scanf(" %c", &a);

int c = 0;
int i = 0;

while (i < strlen(s)) {
    if (s[i] == a) {
            c++;
    }
    i++;
}

printf("the character is present %d times\n", c);
return 0;
}

```

```
enter a string: goodnight
enter the character you want to count: o
the character is present 2 times
```

Task 3: Lowercase to Uppercase

```c
#include <stdio.h>
#include <string.h>

int main () {

char s[200];
printf("enter a string: ");
scanf("%s", &s);

for (int i = 0; i < strlen(s); i++) {
    if (s[i] >= 'a' && s[i] <= 'z') {
        s[i] = s[i] - 'a' + 'A';
    }
}

printf("the uppercase string is %s\n", s);
return 0;
}
```

```
enter a string: codeblocks
the uppercase string is CODEBLOCKS

Process returned 0 (0x0)   execution time : 13.110 s
Press any key to continue.
```

Task 4: Anagram Check

```c
#include <stdio.h>
#include <string.h>

int main () {
char s1[100];
char s2[100];

printf("enter two strings: \n");
scanf("%s%s", &s1, &s2);

char t;

for (int i = 0; i < strlen(s1)-1; i++) {
    for (int j = 0; j < strlen(s1)-i-1; j++) {
        if (s1[j] > s1[j+1]) {
            t = s1[j];
            s1[j] = s1[j+1];
            s1[j+1] = t;
        }
    }
}

for (int i = 0; i < strlen(s2)-1; i++) {
    for (int j = 0; j < strlen(s2)-i-1; j++) {
        if (s2[j] > s2[j+1]) {
            t = s2[j];
            s2[j] = s2[j+1];
            s2[j+1] = t;
        }
    }
}


if (strcmp(s1, s2) == 0) {
    printf("the strings are anagrams");
}

else printf("the strings are not anagrams");
return 0;

}
```

```
enter two strings:
apple
paple
the strings are anagrams
Process returned 0 (0x0)   execution time : 7.105 s
Press any key to continue.
```

Task 5: Remove all Spaces

```c
#include <stdio.h>
#include <string.h>

int main () {
char s[200];
printf("enter a string: ");
fgets(s, sizeof(s), stdin);

int c = 0;
for (int i = 0; i < strlen(s); i++) {
    if (s[i] != ' ') {
        s[c++] = s[i];
    }
}

s[c] = '\0';

printf("the string without spaces is %s", s);
return 0;
}
```

```
enter a string: this is a sentence without spaces
the string without spaces is thisisasentencewithoutspaces

Process returned 0 (0x0)   execution time : 5.806 s
Press any key to continue.
```